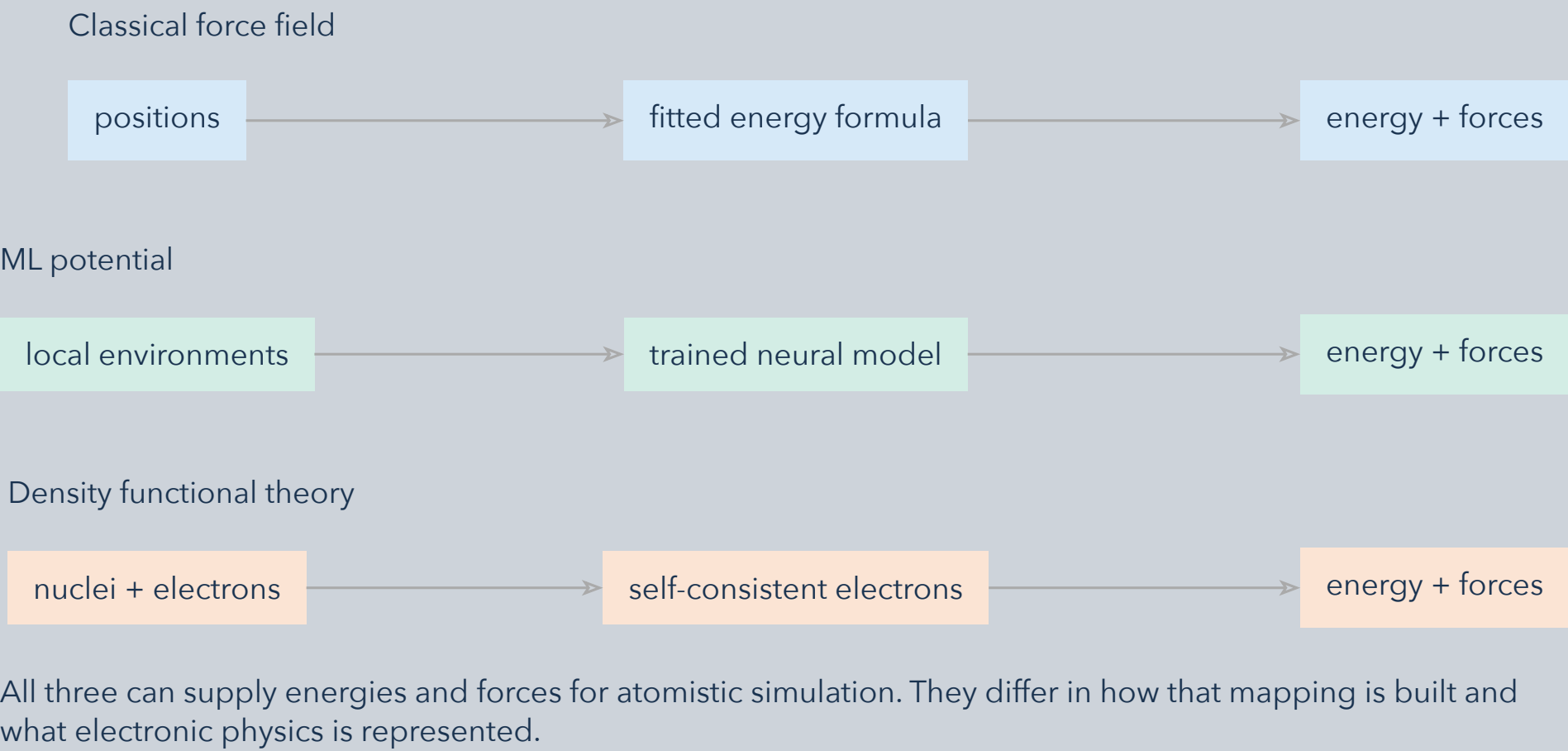


Choose a model for the quantities and configurations your simulation must predict. Speed, accuracy, and transferability depend on the task; they are not universal scores.

## 1 What is evaluated?



## 2 What must be validated?

| Method                | Checks for the intended application                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Classical force field | Functional form and parameter coverage; bonding changes, charges, and long-range interactions.    |
| ML potential          | Training-domain coverage; errors on representative structures, forces, defects, and trajectories. |
| DFT                   | Functional, basis and sampling convergence; relevant charge, spin, and electronic states.         |

Benchmark the same observable, structures, and accuracy target on the intended hardware. Report throughput and errors with units; test transfer on configurations excluded from fitting.

**Ask “good enough for this task?” before “which is best?”** Classical and ML models amortize a fitted approximation over many evaluations. DFT solves an approximate electronic problem for each configuration. None guarantees accuracy outside its validated setting.